



Why science teachers
should not be given
playground duty.

Review for Midterm Exam

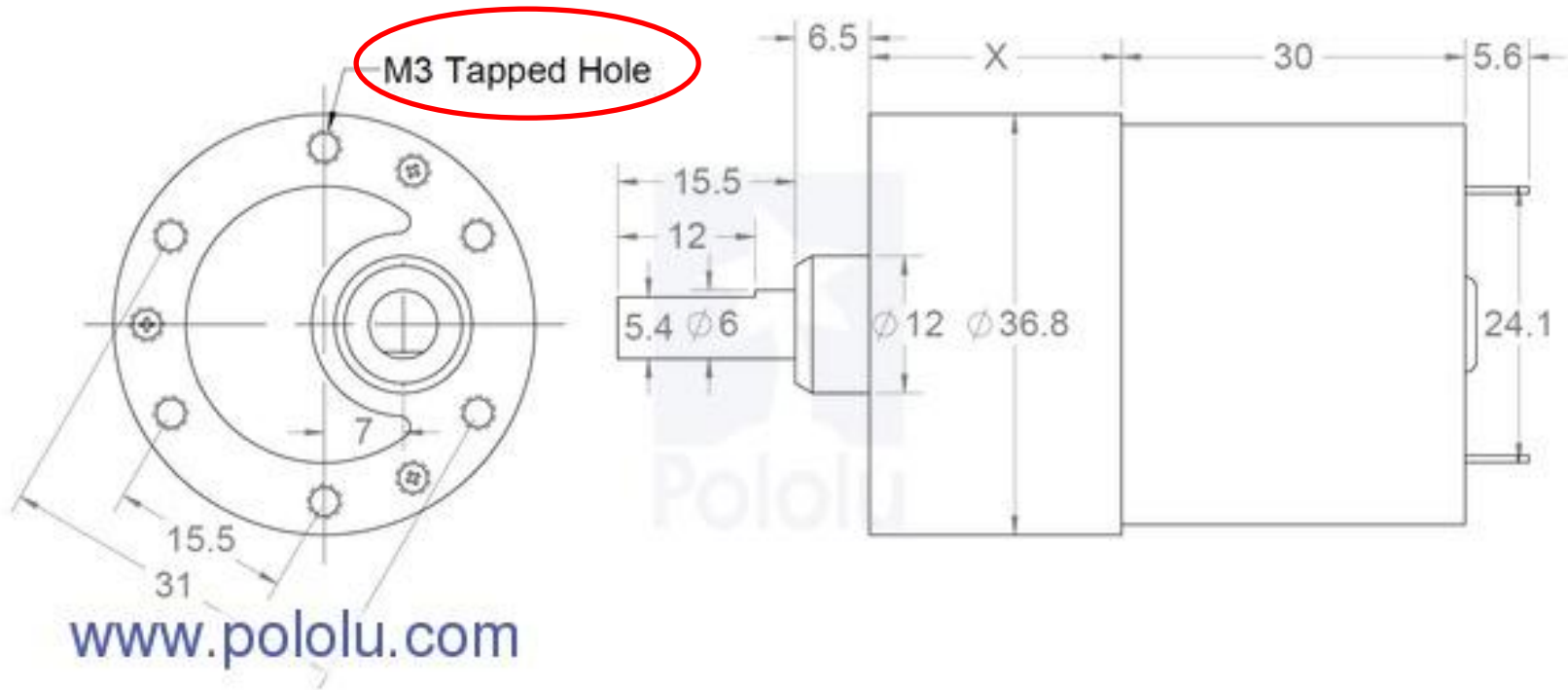
ME 350 Lecture

Sample Parts to Pass Around

ME350 kit motor – with the housing deformed due to incorrect mounting (This motor was clamped in place with a U-bolt instead of mounting to the face of the motor with M3 screws.)



From the Pololu website for the motor



“Warning: Do not screw too far into the mounting holes as the screws can hit the gears. We recommend screwing no further than 3mm (1/8”) into the screw hole.”

Note: M3 x 10mm and M3 x 16mm screws are available in the X50 Assembly Room.

Reminders/Announcements

- Use the signup system when you need to reserve a mill or lathe in the machine shop.
 - **Make sure your approval packages have been approved before signing up to use a machine.**
- **Shoulder bolts** and other fasteners are available in the drawers near the southeast door of the X50 Assembly Room. 1/4" diameter shoulder bolts are available in lengths ranging from 1/2" to 1-1/4".
 - If you need other lengths, you can order them from McMaster.com or go to a hardware store.

Midterm Exam: Logistics

- **Date and Time: Tuesday Sep 30th, during the normal lecture time**
- **Exam will start promptly at 1:30pm**
- **Locations:**
 - **If your last name is from Adegbite to Parikh: This room (Chrysler Auditorium)**
 - **If your last name is from Park to Zou: room 1670 Beyster**
- **Students with accommodations from the SSD office will take the exam in the Mechatronics Lab, starting at the same time.**

Exam Content

- 4-6 problems about linkages and motors. The motor problem could also include a simple gear ratio.
- Each problem could have several parts.
- Exam will **not** include Planetary Gears.
- This exam WILL NOT include Support/Rebut questions.
- Practice problems will be posted on Canvas under Files, Exams.
- **Solutions for HW 1 and 2** have been posted to Canvas. Solution for HW3 will be posted to Canvas on Saturday.

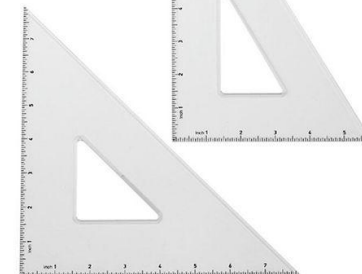
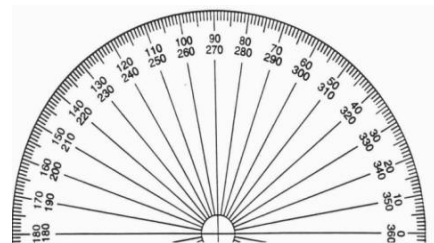
What to Bring to the Exam

- Pencils and Erasers
- Calculator
- Triangles
- Protractor
- Ruler or scale
- Compass
- **One page (double-sided) of handwritten notes (equations, etc.)**



The exam is closed-book. We will provide any tables that are needed. We will not provide formulas.

Turn off cell phones and computers. No devices are allowed that can communicate with others.



Midterm Exam

- You must sign the honor code for exam to be graded.
- To study for the exam:
 - Review the homework problems and solutions posted on Canvas
 - Review the lecture notes, including the example problems
 - Review the practice midterm exam, posted on Canvas. These questions provide an estimate of the level of difficulty that you can expect to see on the exam. They are not intended to be a complete list of all topics that could be covered.
- During the exam:
 - Don't get stuck on any one problem. Pace yourself, and attempt to solve all the problems, to maximize partial credit!



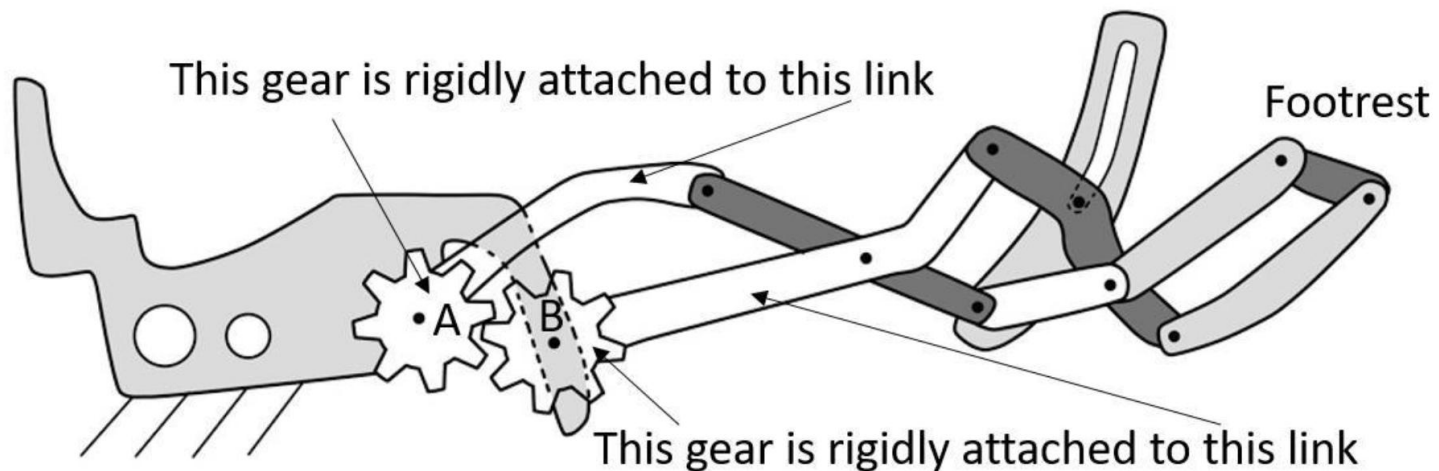
Practice Problems

Problem 1: Mobility

The figure below shows the footrest portion of a recliner mechanism. The leftmost link is attached to the ground and does not move. How many actuators are needed to extend the footrest if:

- (a) The mechanism is as shown above?
- (b) The gears are removed from the mechanism, leaving only the pin joints at points A and B?

State all assumptions and show your work clearly.



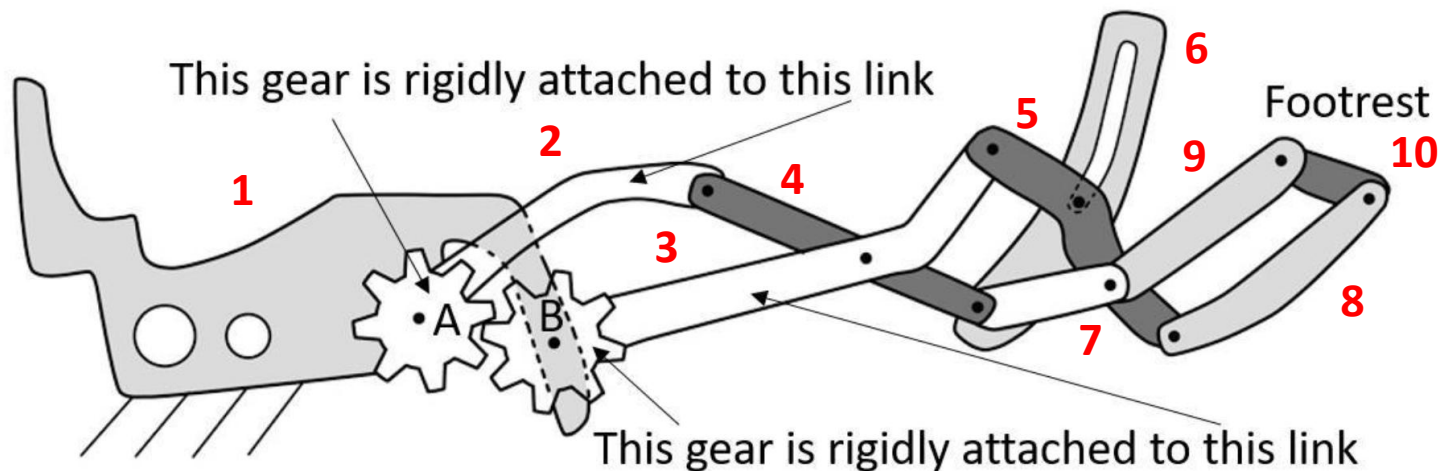
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State all assumptions and show your work clearly.

Red = # Links



Problem 1: Mobility

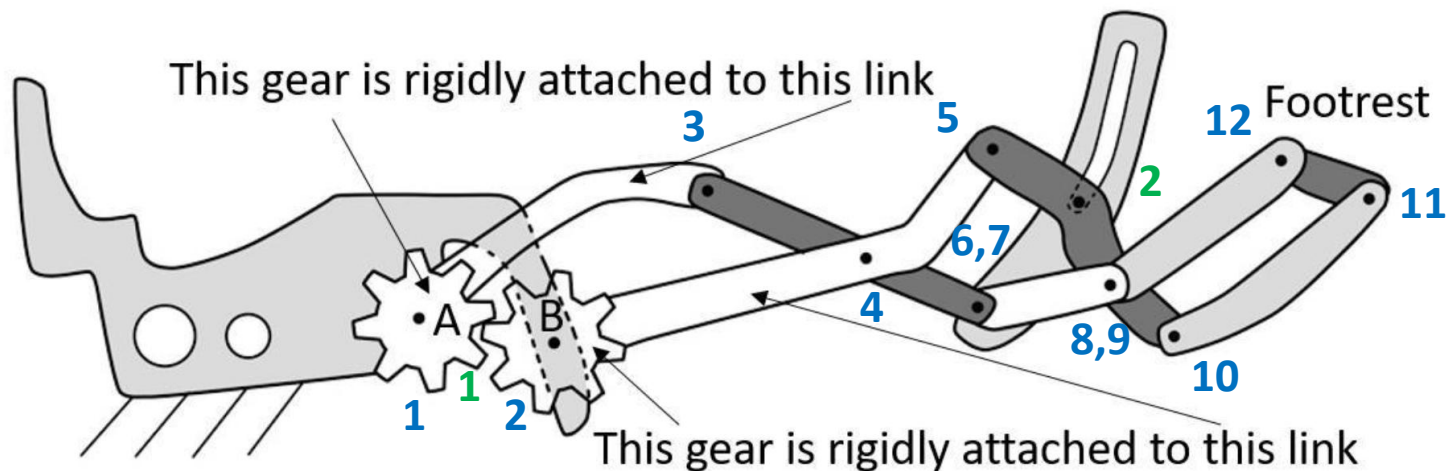
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- (a) The mechanism is as shown above?
- (b) The gears are removed from the mechanism, leaving only the pin joints at points A and B?

Blue = # J1

Green = # J2

State all assumptions and show your work clearly.



Solution to Problem 1

Solution:

4 points for having the correct Grashof formula ($M = 3(L-1) - 2*J1 - J2$)

(a) The mechanism is as shown above (including the gears)?

2 points correct links (L: 10)

2 points correct J1 (J1: 12)

2 points correct J2 (J2: 2) (pin in slot, and gear joint)

2 points correct answer (M: 1)

[1 pt if the student has the incorrect answer, but notes that answer is intuitively wrong]

(b) The gears are removed from the mechanism, leaving the pin joints at points A and B

2 point correct Links (L: 10)

2 points correct J1 (J1: 12)

2 points correct J2 (J2: 1)

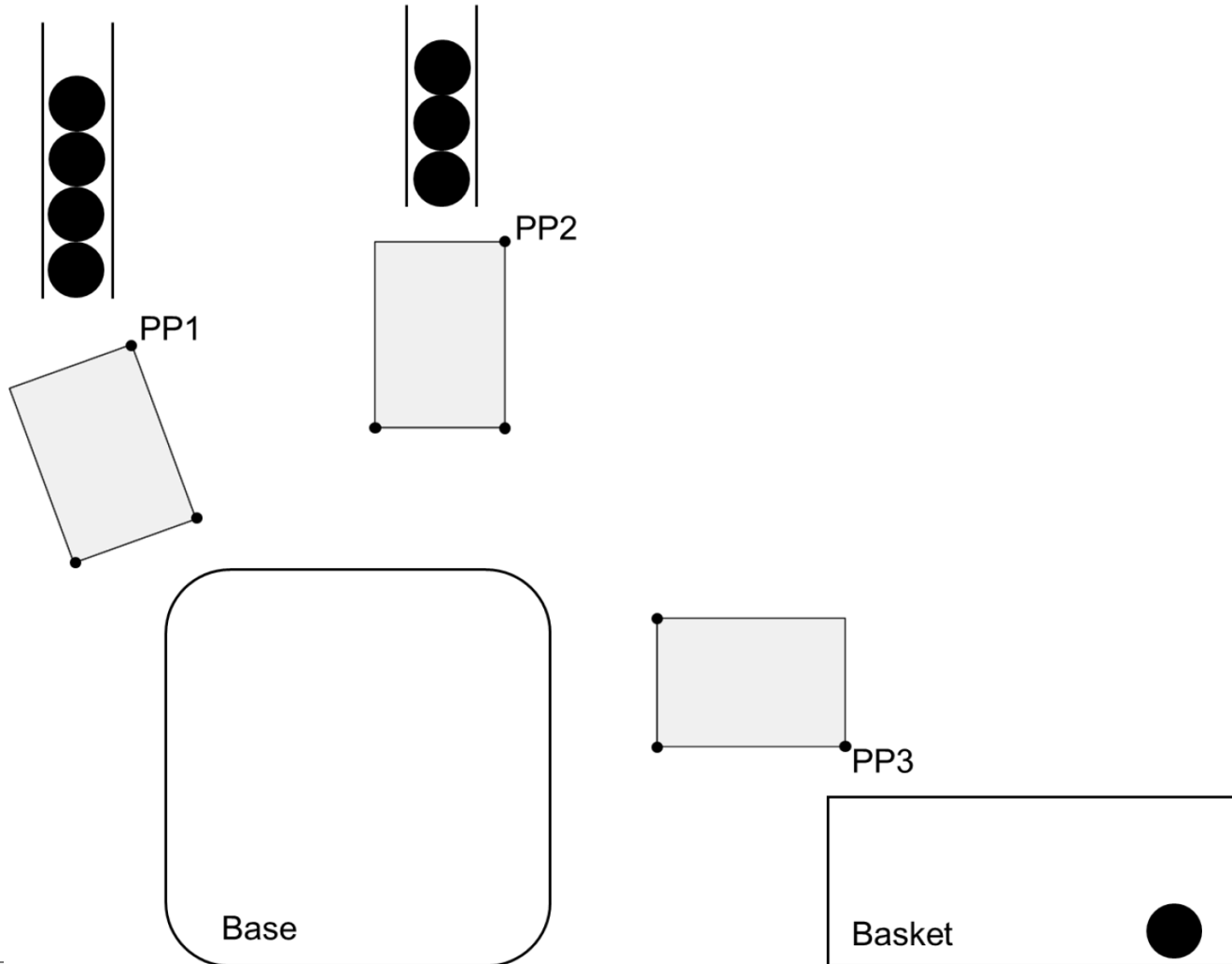
2 points correct answer (M: 2)

[1 pt if the student has the incorrect answer, but notes that answer is intuitively wrong]

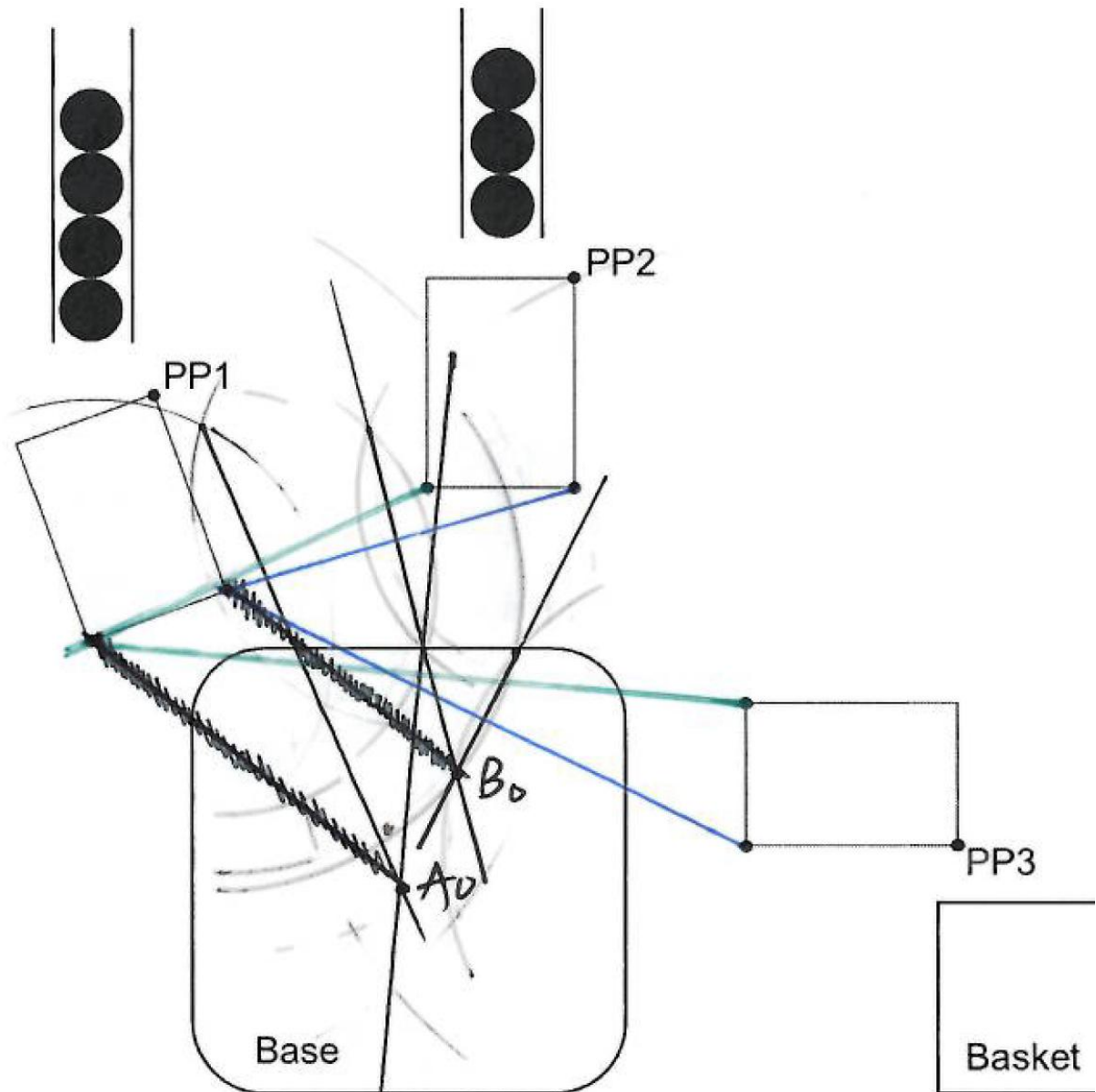


Graphical Synthesis

Use the graphical method to design a four-bar linkage to move the cup in planar motion from PP1 to PP2 to PP3 in the orientation shown. Draw the four-bar linkage in position 1. *The input and output links can be in a different plane. The ground pivots can be anywhere in the base but the moving pivots should be located on the black circles on the bottom edge of cup.*



Solution to Graphical Synthesis -- Need to specify:



Analytical Synthesis

On this page you will set up the same problem for solution using the analytical method. Note: You do not need to use the same ground pivots and moving pivots as part I, and your solution does not need to be feasible.

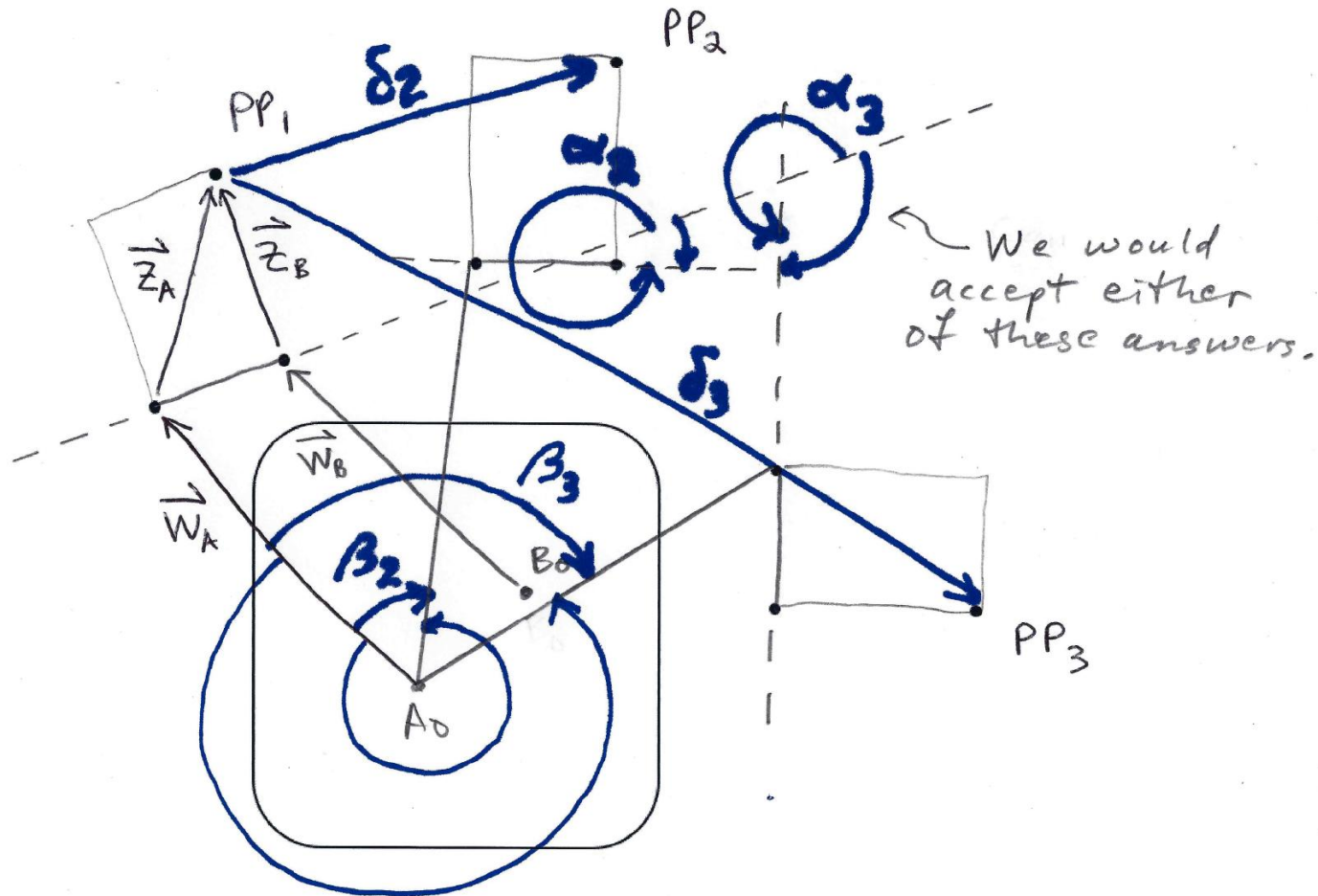
- II. Draw and label the δ_j 's and the α_j 's in the standard dyad form. It must make use of the precision points given.
- III. In position 1, draw and label the **W's** and the **Z's** in the standard dyad form. It must make use of the precision points given. Do not attempt to solve the problem.
- IV. Draw the same four-bar linkage in position 2 (the middle position) and position 3. Draw and label the β 's in standard dyad form.
- V. Based on your drawing for part (II)-(IV), identify what quantities are known (given), what are unknowns, and what are free choices in the standard form for a 3-point motion generation problem.

NOTE: In this exam problem, you were only asked to draw and label the vectors and angles.

In the midterm exam, you may also be given a linkage that has already been designed, and asked to measure and report the values of the vectors (W, Z, and deltas in cartesian coordinates) and angles (alphas and betas) of the linkage in particular positions.



Solution to Analytical Synthesis



Knowns: $\alpha_2, \alpha_3, \vec{\delta}_2, \vec{\delta}_3$

Unknowns: $\vec{Z}, \vec{W}$

Free Choices: β_2, β_3



STRETCH!



Image: Doga Kurt (cat: Biscuit)

Motors Practice Question

You have been asked to select a permanent magnet brushed DC motor for a client's application. The application uses the motor to fetch oil in a pan from a well using a 18 V power supply. The motor experiences load torque $T_{L1} = A + B\omega$ in Part 1 of the motion when the pan is moving through oil, and $T_{L2} = A$ in Part 2 of the motion when the pan is moving through air; where $A = 2$ Nm, $B = 0.02$ Nm/rpm, and ω is the speed of the motor in rpm. The client has the following criteria:

- I) The motor should rotate as fast as possible in Parts 1 and 2 of the motion.
- II) The motor should experience minimal change in speed when transitioning from Part 1 to Part 2 of the motion.

To satisfy Criterion I, you choose to operate the motor at the **maximum voltage** throughout the motion:

- (a) Which of the motors in the table below is most suitable to satisfy Criterion II above? In one sentence, justify your selection.
 - Table is on next slide
- (b) Using the motor you selected in (a) above, calculate ω_1 and ω_2 , the values of ω in Parts 1 and 2 of the motion, respectively.

The client is unhappy with the change in speed from ω_1 to ω_2 . So, you decide to operate the motor with different supply voltages in order to maintain the **same speed** in Parts 1 and 2 of the motion.

- (c) Which of the motors in the table below is most suitable to satisfy Criterion I above? In one sentence, justify your selection.
 - Table is on next slide
- (d) Using the motor you selected in (c) above, calculate the supply voltages V_{s1} and V_{s2} for Parts 1 and 2 of the motion, respectively.

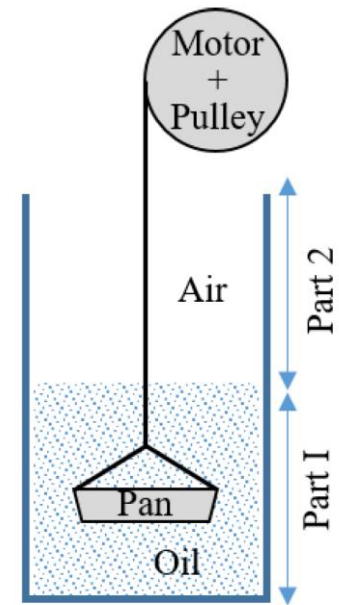


Table for the Motors Practice Question

Motor #	1	2	3
Nominal Voltage (V)	6	9	6
Stall Torque (Nm)	2	3	2.5
Torque-speed gradient (Nm/rpm)	0.04	0.05	0.04



Motors: Solution

PROBLEM 4: Version 1

The letters A, B, C, etc., are entered into Canvas to let you know what portions you got right. Any letter that is not entered in Canvas means that you did not get that portion right (or did not have that portion) in your solution.

Note Title

9/27/2020

a) Motor 2 will best satisfy Criterion II because it has highest K

[A] 2 pts for correct choice AND reason. (No point for correct motor with wrong reason)

b) $T_s = \frac{V_s}{V_{s,nom}} \cdot T_{s,nom} \therefore T_s - K\omega = A + B\omega$

[C] 2 pts for equating motor torque to load torque for Part 1

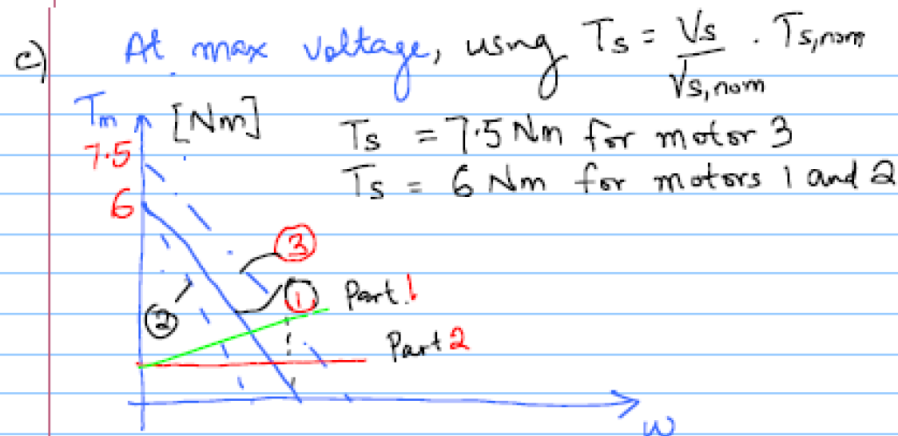
$$\Rightarrow \omega_1 = \frac{T_s - A}{K + B}; \quad \omega_2 = \frac{T_s - A}{K}$$

[B] 2 pts converting from nominal stall torque to stall torque at supply voltage

[D] 2 pts for equating motor torque to load torque for Part 2

For motor 2, $T_s = \frac{18}{9} \times 3 = 6 \text{ Nm}$; $\omega_1 = \frac{6 - 2}{0.05 + 0.02} = 57.14 \text{ rpm}$; $\omega_2 = \frac{6 - 2}{0.05} = 80 \text{ rpm}$

[E] 1 pts each for correct speeds (2 pts total)



Motor 3 is best because it provides the highest speeds in Parts 1 and 2 because it has highest $\omega_{max} = T_s/K$ (see figure)

[F] 2 pts for correct motor AND reason. (No points for correct motor with wrong reason)



d) From the figure above, Part 1 is more critical than Part 2 in terms of voltage (torque) requirement. Therefore, use the max. voltage for Part 1

[G] 2 pts for selecting max. supply voltage for Part 1

$$\boxed{V_{s1} = 18 \text{ V}} \Rightarrow \omega_1 = \frac{T_s - A}{K + B} = \frac{7.5 - 2}{0.04 + 0.02} = 91.67 \text{ rpm} = \omega_2$$

[H] 2 pts for equating ω_1 to ω_2

For Part 2:

[I] 2 pts for determining T_s for Part 2

$$T_s = A + K\omega_2 = 5.67 \text{ Nm}$$

$$\frac{V_s}{T_s} = \frac{V_{s, \text{nom}}}{T_{s, \text{nom}}} \Rightarrow$$

$$\boxed{V_{s,2} = \frac{6}{2.5} \times 5.67 = 13.6 \text{ V}}$$

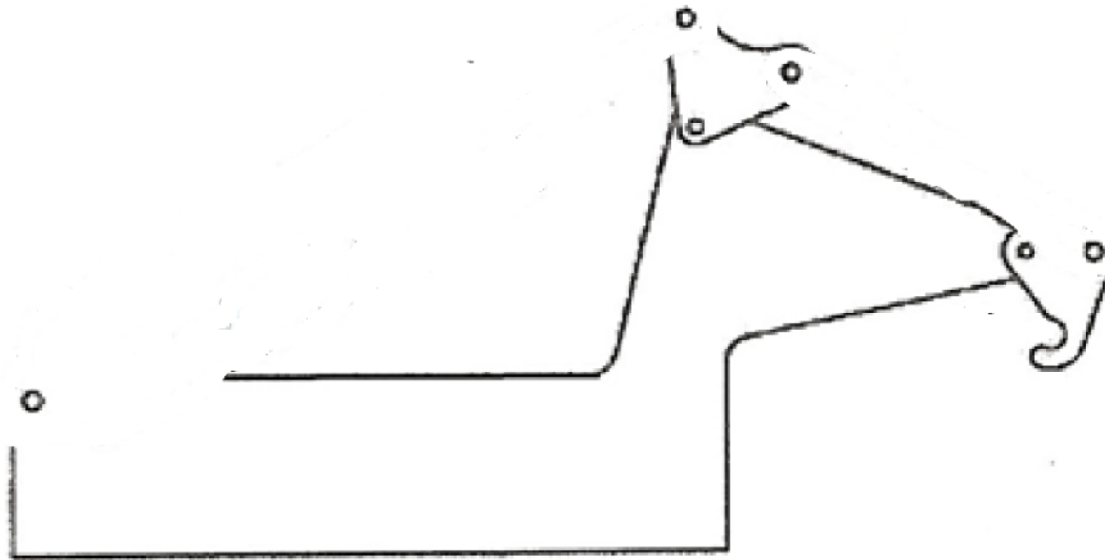
[J] 2 pts for correct answer (voltage) for Part 2



Partial Image from the Answer Sheet

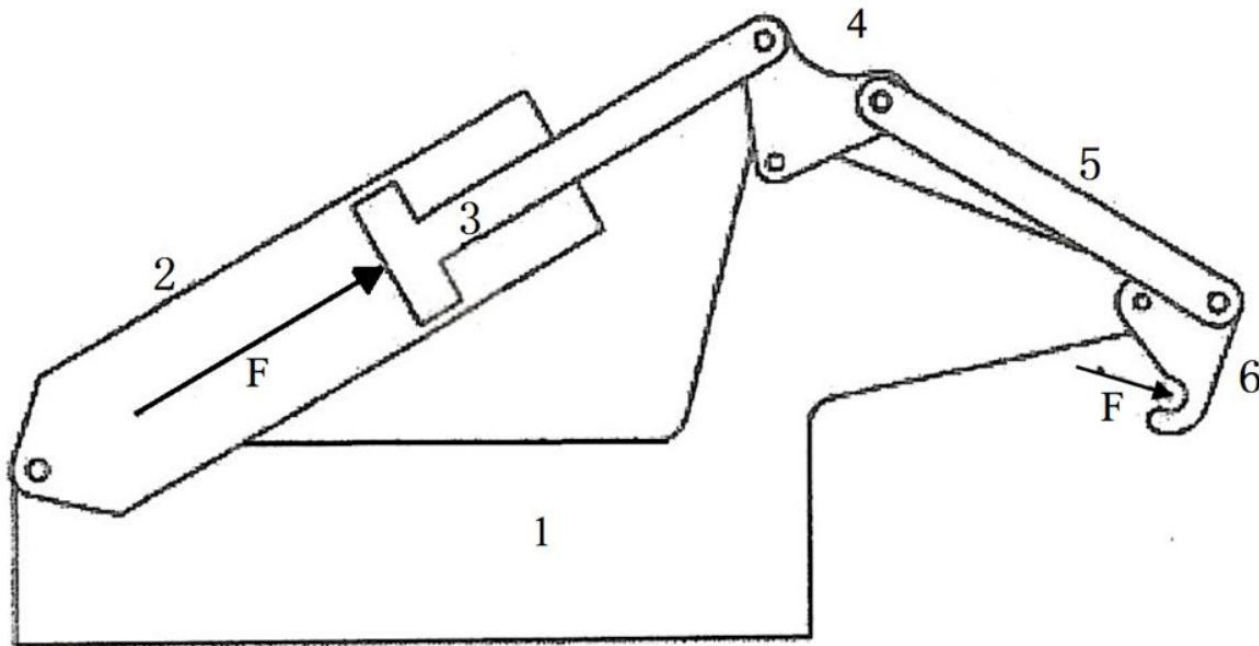
Problem 5

The figure below is incomplete. The complete figure will be provided during the exam.



Mechanical Advantage

For the linkage shown below, the links are numbered, and the input force (on the piston) and resisting force (on the output link 6) are shown with arrows. Calculate the mechanical advantage in the position shown. Use instant centers to do this.

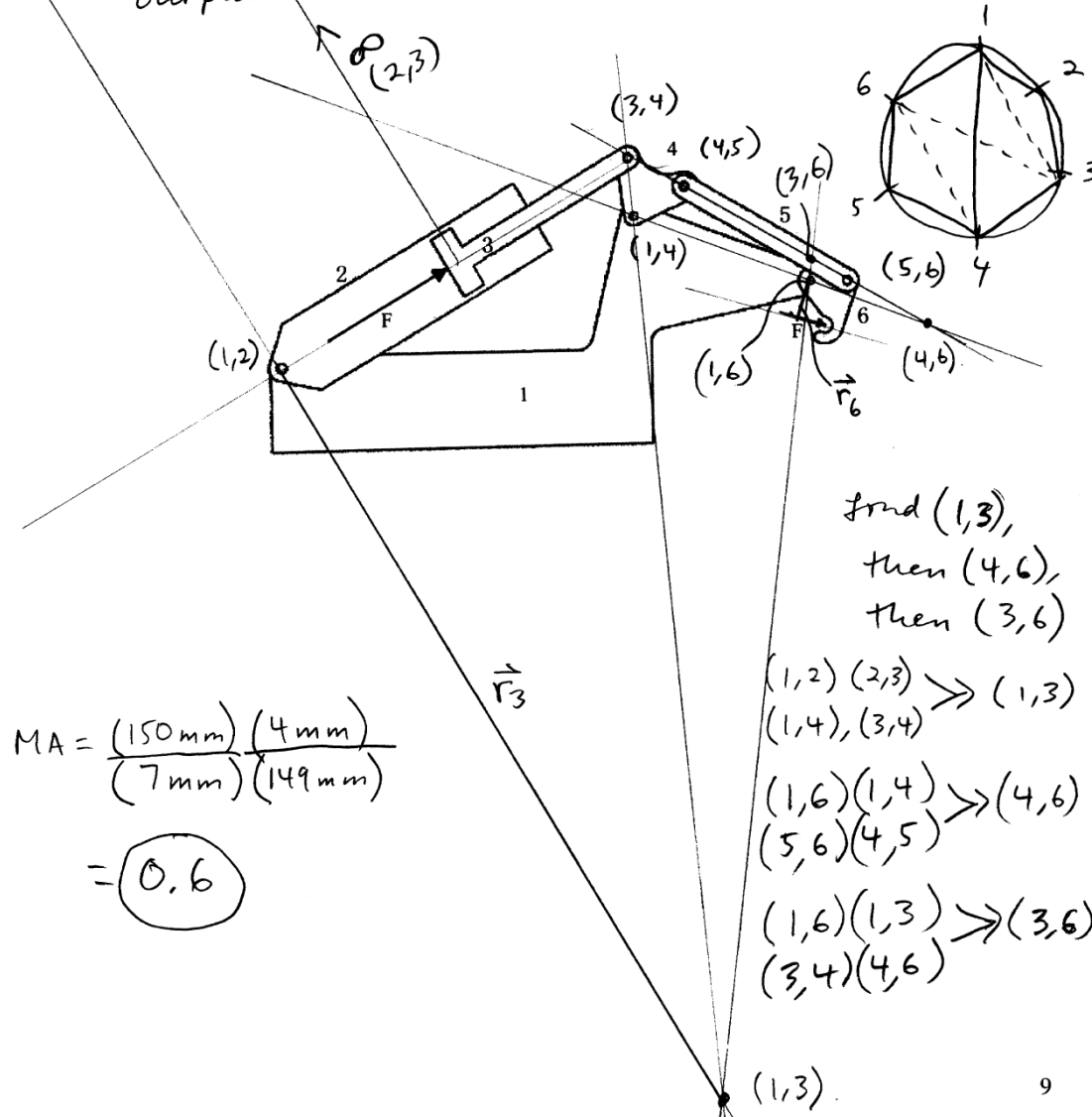


Solution

For the linkage shown below, the links are numbered, and the input force (on the piston) and resisting force (on the output link 6) are shown with arrows. Calculate the mechanical advantage in the position shown. Use instant centers to do this.

Input = link 3
output = link 6.

$$MA = \frac{r_3 \perp F_{in}}{r_6 \perp F_{out}} = \frac{(1,6 - 3,6)}{(1,3 - 3,6)}$$



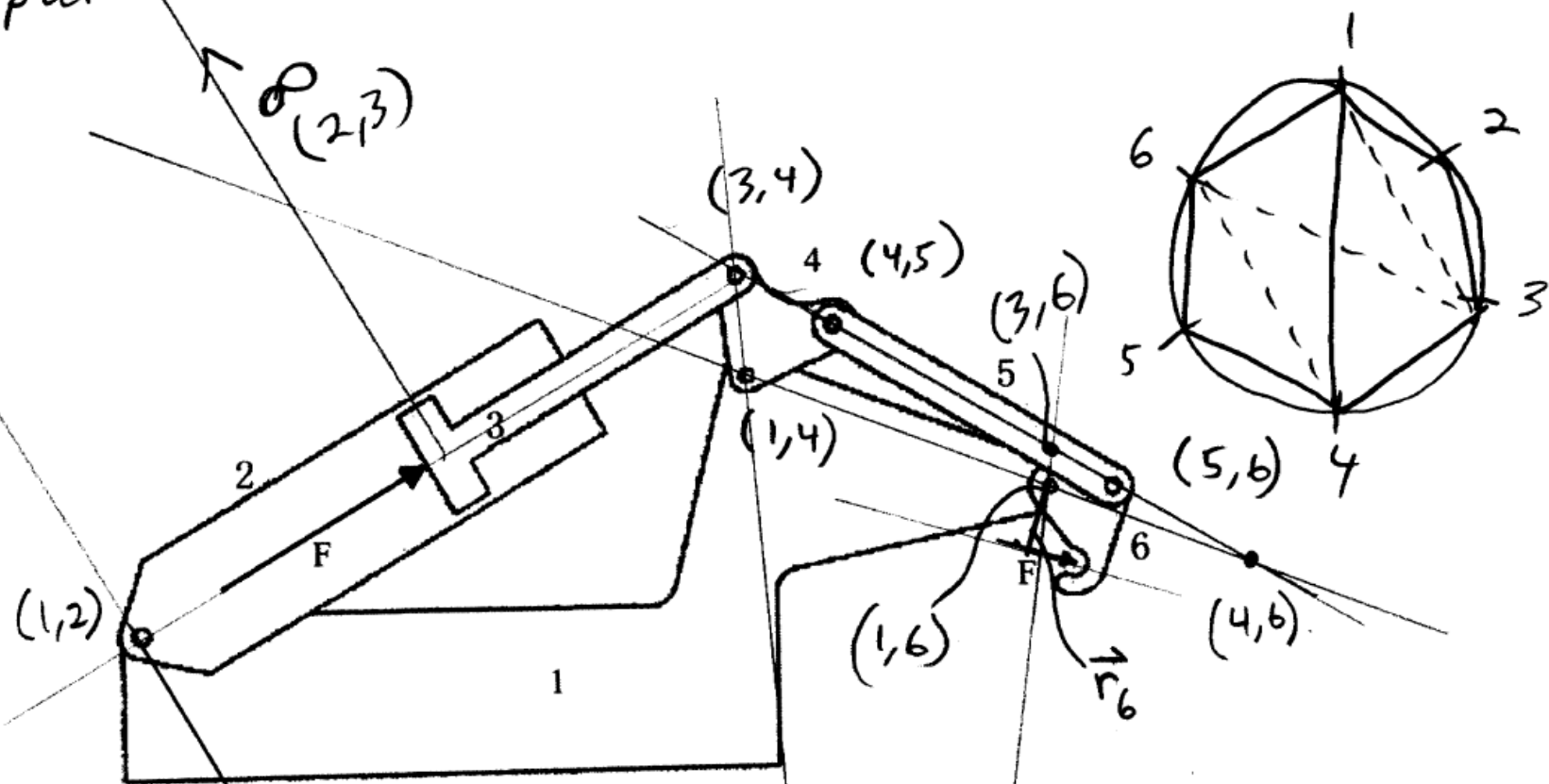
$$MA = \frac{(150 \text{ mm}) (4 \text{ mm})}{(7 \text{ mm}) (149 \text{ mm})}$$

$$= 0.6$$

For the linkage shown below, the links are numbered, and the input force (on the piston) and resisting force (on the output link 6) are shown with arrows. Calculate the mechanical advantage in the position shown. Use instant centers to do this.

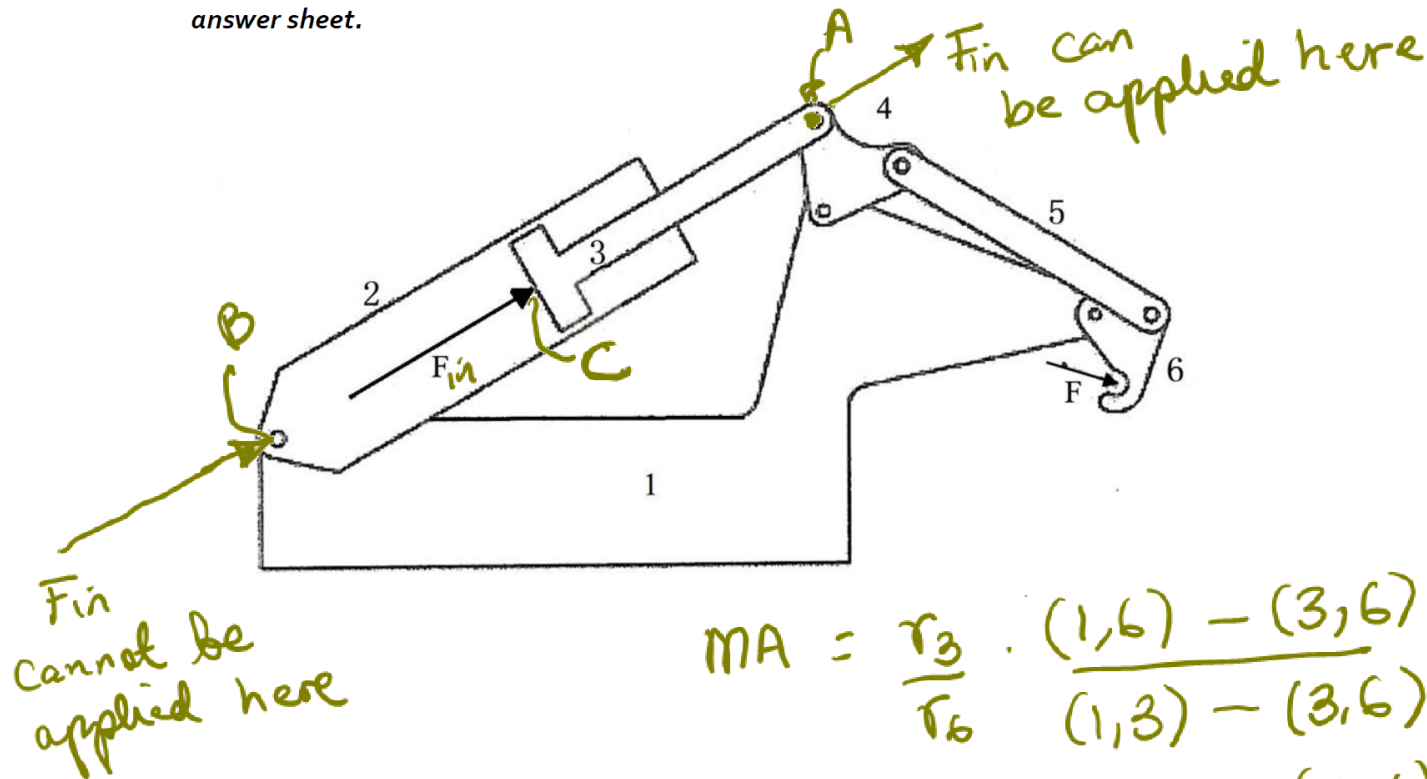
Input = link 3
output = link 6.

$$MA = \frac{r_3^{\perp F_{in}}}{r_6^{\perp F_{out}}} = \frac{(1,6 - 3,6)}{(1,3 - 3,6)}$$



Find (1,3),
then (4,6),
then (3,6)

For the linkage shown below, the links are numbered, and the input force (on the piston) and resisting force (on the output link 6) are shown with arrows. Calculate the mechanical advantage in the position shown. Use instant centers to do this. *Upload your solution worked on the provided answer sheet. Sketch in the portions of the figure missing in the answer sheet.*



$$MA = \frac{r_3}{r_6} \cdot \frac{(1,6) - (3,6)}{(1,3) - (3,6)} \quad \begin{matrix} \checkmark \\ \text{(Correct} \\ \text{+ harder)} \end{matrix}$$

$$MA = \frac{r_4}{r_6} \cdot \frac{(1,6) - (4,6)}{(1,4) - (4,6)} \quad \begin{matrix} \checkmark \\ \text{(Correct} \\ \text{+ easier)} \end{matrix}$$

$$MA = \frac{r_2}{r_6} \cdot \frac{(1,6) - (2,6)}{(1,2) - (2,6)} \quad \begin{matrix} \times \\ \text{(incorrect)} \end{matrix}$$

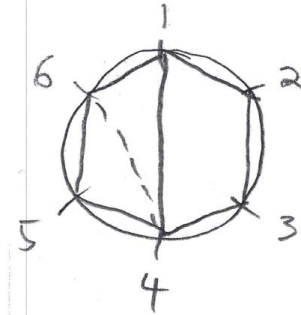
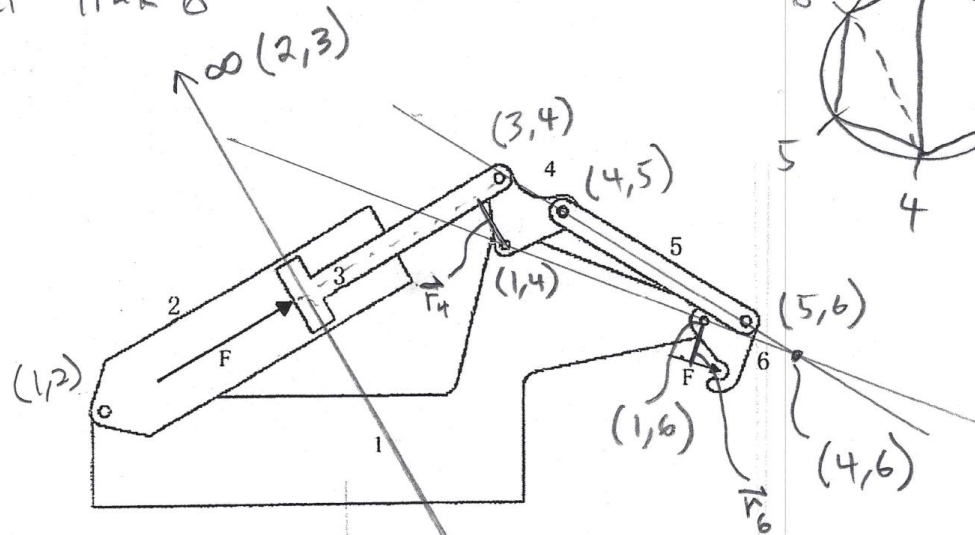
Point A on link 4 can be used as an alternative point for Fin because it has the same velocity in the direction of Fin as point C. It is NOT because it lies on the line of action of Fin.

Point B on link 2 lies on the line of action of Fin but is not an accurate alternative because it does not have the same velocity in the direction of Fin as point C.



Alternate Solution (using link 4 as the input)

Input: link 4.
Output: link 6



$$MA = \frac{\vec{r}_4 (1,6 - 4,6)}{\vec{r}_6 (1,4 - 4,6)}$$

$$\begin{matrix} (1,6)(1,4) \\ (4,5)(5,6) \end{matrix} \rightarrow (4,6)$$

$$MA = \frac{(9 \text{ mm})(15 \text{ mm})}{(7 \text{ mm})(48 \text{ mm})}$$

$$= 0.4$$

The difference is due to measurement error.

